

FLIPPER BLACK

Component Research & Datasheet Guide

Version 2.0 · Corrected May 2026 · Solo build reference

This guide gives you every official datasheet, LCSC/Mouser part number, key specs to verify, errata to watch for, and search strategies for every component in the Flipper Black BOM. Read each chip section before ordering and before placing it in KiCad.

■ RTL-SDR is NOT integrated on the PCB. It connects as an external USB dongle to a laptop running SDR# or GQRX. See Section 8 for the corrected architecture.

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1. How to Read a Datasheet

A datasheet is the manufacturer's official specification document for a chip. Most beginners skip datasheets and regret it. Here is the correct reading order:

Step 1 — First page

The one-line description confirms you have the right chip. For example: 'CC1101: Low-Power Sub-1GHz RF Transceiver.' If unexpected, stop — wrong datasheet.

Step 2 — Features list

Usually on page 1–2. Check frequency range, interface type (SPI/I2C/UART), voltage range, and any feature you are counting on. If missing from this list, the chip does not have it.

Step 3 — Pin description table

Lists every pin by name, number, type, and function. The most-used page. You will return here constantly when drawing your schematic.

Step 4 — Absolute maximum ratings

Limits you must NEVER exceed — max voltage, max current, max temperature. Exceeding any of these kills the chip immediately or eventually.

Step 5 — Typical application circuit

Most datasheets include a reference schematic. Copy it exactly first, then modify only what you need to. Always start here.

Step 6 — Register map

For configurable chips (CC1101, SX1276). Shows each register's address, bit fields, and default value. Used when writing driver code.

Step 7 — Errata

Separate documents listing known silicon bugs. Always download and read the errata before finalizing your design.

Key datasheet terminology

Term	Meaning
VCC / VDD	Positive supply voltage (power input to chip)
GND / VSS	Ground reference (0 V return path)
I/O voltage	Voltage level for logic signals (usually 3.3 V)
SPI / I2C / UART	Communication interface type
MOSI / MISO / SCK / CS	SPI bus signal names
Active LOW	Pin does its job when pulled to 0 V (written as /CS or CS#)
Pull-up / pull-down	Resistor holding a pin at VCC or GND when floating
Decoupling cap	100 nF ceramic cap on every VCC pin — mandatory

Term	Meaning
QFN	Quad flat no-lead package — pads on bottom, needs reflow soldering
SOIC	Gull-wing legs — hand-solderable with a fine tip
Errata	Official list of known bugs in the chip silicon

2. ESP32-S3 — Main MCU

Document	URL
ESP32-S3 Datasheet	espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf
Technical Reference Manual	espressif.com/sites/default/files/documentation/esp32-s3_technical_reference_manual_en.pdf
Hardware Design Guidelines	docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s3/
Errata	espressif.com/sites/default/files/documentation/esp32-s3_errata_en.pdf
ESP-IDF Programming Guide	docs.espressif.com/projects/esp-idf/en/stable/esp32s3/
Schematic Checklist	docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s3/schematic-checklist.html

Key specs

Spec	Value	Why it matters
CPU cores	2x Xtensa LX7	Confirms dual-core for parallel tasks
CPU speed	240 MHz max	Max clock for SPI and firmware timing
Supply voltage	3.0–3.6 V	Your LDO must output 3.3 V $\pm$ 0.3 V
GPIO count	45 usable	Plan pin assignment early — you will run short
USB OTG	Full-speed USB 2.0	Required for RTL-SDR external dongle connection
I/O voltage	3.3 V (NOT 5 V tolerant)	Never connect 5 V signals directly
Crystal frequency	40 MHz required	Buy 40 MHz crystal — NOT 26 MHz, NOT 32 MHz
Strapping pins	GPIO0, GPIO3, GPIO45, GPIO46	Must be correct at boot or chip fails to start

Required external components

Component	Value / Part	LCSC #	Purpose
40 MHz crystal	40 MHz, 10 pF load, SMD 3225	C256661	Clock source — chip won't run without it
Crystal load caps	10 pF, 0402, C0G/NP0	C14858	Matched to crystal spec — use exact value
SPI flash	W25Q128JVS1Q, WSON-8	C97521	Firmware storage — 16 MB
Core decoupling	100 nF per VCC pin, 0402	C14663	One per each VDD pin
Bulk cap	10 μ F, 0603, X5R	C19702	Handles RF TX current surge
CHIP_PU pull-up	10 kOhm, 0402	C98220	Enables chip — float = no boot
Reset button	Tactile SMD	C318884	Pull CHIP_PU to GND to reset
GPIO0 pull-up	10 kOhm, 0402	C98220	HIGH = normal boot, LOW = download mode

Component	Value / Part	LCSC #	Purpose
WiFi/BLE antenna	2.4 GHz PCB trace or U.FL	C9002	Required for WiFi and BLE to function

✗ **CRITICAL:** The ESP32-S3 QFN56 has an exposed thermal pad on the bottom. It must be soldered to a large ground pad on the PCB with multiple thermal vias. Floating thermal pad = overheating and poor RF performance.

✓ **TIP:** Download the Hardware Design Guidelines PDF and follow the schematic checklist before submitting Gerbers. Espressif's checklist catches 90% of first-timer mistakes.

3. W25Q128 — External SPI Flash

Document	URL
W25Q128JV Datasheet	winbond.com/resource-files/w25q128jv_dth_rev06_11162016_plus.pdf

Key specs

Spec	Value
Capacity	128 Mbit = 16 MB
Interface	Standard / Dual / Quad SPI
VCC range	2.7–3.6 V
Max clock (QIO)	133 MHz
Erase/write cycles	100,000 min
Data retention	20 years min
JEDEC Manufacturer ID	0xEF (Winbond)
Device ID (128Mb)	0x4018 — verify at boot with <code>esptool.py flash_id</code>

Pin wiring to ESP32-S3

W25Q128 Pin	Pin Name	Connect to	Notes
1	/CS	ESP32 GPIO10	Active LOW chip select
2	DO (IO1)	ESP32 GPIO13	MISO in single SPI, IO1 in QSPI
3	/WP (IO2)	ESP32 GPIO14	Pull HIGH via 10 kOhm if not using QSPI
4	GND	GND plane	Via to bottom ground plane
5	DI (IO0)	ESP32 GPIO11	MOSI in single SPI, IO0 in QSPI
6	CLK	ESP32 GPIO12	Clock from ESP32
7	/HOLD (IO3)	ESP32 GPIO15	Pull HIGH via 10 kOhm if not using QSPI
8	VCC	ESP32 VDD_SPI	100 nF + 1 uF decoupling caps right here

✓ TIP: Run `'esptool.py flash_id'` after soldering to verify chip responds with Manufacturer: 0xEF, Device: 0x4018. If you see 0xFF, there is a bad solder joint on one of the 6 signal pins.

4. CC1101 — Sub-GHz Transceiver

Document	URL
CC1101 Datasheet	ti.com/lit/ds/symlink/cc1101.pdf
CC1101 User's Guide	ti.com/lit/ug/swra173b/swra173b.pdf
DN013 — RF circuit design	ti.com/lit/an/swra161b/swra161b.pdf
SmartRF Studio 7 (free tool)	ti.com/tool/SMARTRFSTUDIO
rtl_433_ESP library	github.com/NorthernMan54/rtl_433_ESP

Key specs

Spec	Value
Frequency range	300–928 MHz
Modulation	2-FSK, GFSK, ASK/OOK, 4-FSK, MSK
Interface	SPI (up to 10 MHz)
Supply voltage	1.8–3.6 V
TX power (no PA)	-30 to +10 dBm
RX sensitivity	-112 dBm at 1.2 kbps
Data rate	0.6 to 600 kbps
Package	QFN-20, 4x4 mm
XTAL frequency	26 MHz (external crystal required)

Critical registers

Register	Address	Purpose	What to set
IOCFG0	0x00	GDO0 pin function	0x06 = assert on sync word
FREQ2/1/0	0x0D-0x0F	Operating frequency	Use SmartRF Studio to calculate
MDMCFG4	0x10	Channel bandwidth + data rate	Depends on protocol
MDMCFG2	0x12	Modulation, sync mode	0x03 = 2-FSK, 30/32 sync
PATABLE	0x3E	TX power level	0xC0 = max ~+10 dBm
MCSM1	0x17	State after TX/RX	0x3F = stay in RX after receive

✓ TIP: Use TI SmartRF Studio 7 (free) to generate all CC1101 register values for your target frequency and modulation. Copy the output directly into firmware. Never calculate register values manually.

✗ CRITICAL: For raw CC1101 chip, buy only from authorized distributors (Mouser, DigiKey, LCSC). Counterfeit RF chips are common on AliExpress — they look identical but have completely wrong RF characteristics.

5. SX1276 — LoRa Transceiver

Document	URL
SX1276 Datasheet	semtech.com/uploads/documents/DS_SX1276-7-8-9_W_APP_V7.pdf
Application Note AN1200.13	semtech.com/uploads/documents/an1200.13.pdf
arduino-LoRa library	github.com/sandeepmistry/arduino-LoRa
RadioLib (advanced)	github.com/jgromes/RadioLib

LoRa configuration parameters

Parameter	Range	Effect	Recommendation
Spreading Factor	SF7–SF12	Higher SF = longer range, slower data rate	SF9 for balanced range/speed
Bandwidth	7.8–500 kHz	Wider = faster but noisier reception	125 kHz standard
Coding Rate	4/5–4/8	Higher = more error correction overhead	4/5 normal, 4/8 noisy environment
TX Power	+2 to +20 dBm	Higher = longer range, more current (120 mA at +20 dBm)	+17 dBm for 10 km target
Frequency	137–1020 MHz	Must match receiver. Check regional rules.	868 MHz EU / 915 MHz US

SPI pin connections to ESP32-S3

SX1276 Pin	Function	Connect to ESP32-S3	Notes
NSS	Chip select (active LOW)	Any free GPIO (e.g. GPIO5)	Separate CS pin from CC1101
MOSI	SPI data in	SPI2 MOSI (GPIO11 if shared)	Share SPI bus with CC1101
MISO	SPI data out	SPI2 MISO (GPIO13)	Share SPI bus
SCK	SPI clock	SPI2 SCK (GPIO12)	Share SPI bus
RESET	Reset (active LOW)	Any free GPIO (e.g. GPIO20)	Pull HIGH via 10 kOhm
DIO0	TX done / RX done interrupt	Any GPIO with interrupt (e.g. GPIO21)	MUST connect — do not float
DIO1	RX timeout / FHSS	Optional GPIO	Connect if using FHSS or RX timeout
VCC	3.3 V supply	3.3 V rail	100 nF + 10 uF decoupling
GND	Ground	GND plane	Multiple vias under chip

✗ **CRITICAL:** DIO0 must be connected to a GPIO. When a LoRa transmission completes or a packet arrives, DIO0 pulses HIGH. Without this interrupt pin your firmware must poll constantly and will miss packets.

6. PN532 — NFC / RFID Controller

Document	URL
PN532 User Manual	nxp.com/docs/en/user-guide/141520.pdf
PN532 Application Note (antenna design)	nxp.com/docs/en/application-note/AN10609.pdf
Adafruit PN532 Library	github.com/adafruit/Adafruit-PN532
libnfc (advanced research)	github.com/nfc-tools/libnfc

Interface mode selection — set before PCB is made

The PN532 supports three interface modes, selected by HSU0 and HSU1 pins. You CANNOT change mode in firmware. Decide before routing your PCB.

HSU1	HSU0	Mode	Use for Flipper Black?
GND	GND	UART (HSU)	No — UART pins needed for GPS
GND	VCC	SPI	YES — fastest, most reliable
VCC	GND	I2C	Alternative if SPI pins are congested
VCC	VCC	I2C	Same as above

✓ TIP: Set HSU1=GND and HSU0=VCC by tying the pins directly to the correct rail on the PCB — not via resistors. This permanently selects SPI mode. Do this at schematic time, not later.

7. NEO-6M — GPS Receiver

Document	URL
NEO-6 Hardware Integration Manual	content.u-blox.com/sites/default/files/products/documents/NEO-6_HardwareIntegrationManual_(GPS.G6-HW-09007).pdf
u-blox 6 Receiver Protocol Spec	content.u-blox.com/sites/default/files/products/documents/u-blox6_ReceiverDescriptionProtocolSpec_(GPS.G6-SW-10018).pdf
TinyGPS++ Library	github.com/mikalhart/TinyGPSPlus
u-center (free desktop test tool)	u-blox.com/en/product/u-center

Key wiring

NEO-6M Pin	Connect to	Notes
VCC	3.3 V rail	100 nF + 10 uF decoupling. Draws ~45 mA during acquisition.
GND	GND plane	Short via to bottom plane
TXD (GPS output)	ESP32 UART RX (e.g. GPIO44)	GPS sends NMEA sentences — this is the data you read
RXD (GPS input)	ESP32 UART TX (e.g. GPIO43)	Send UBX config commands to GPS here
TIMEPULSE	Optional GPIO	1 pulse per second when locked — useful for timing
VBAT	3 V coin cell or supercap	Without it every power cycle = cold start (60s). With it = warm start (5s).
RF_IN	Patch antenna via U.FL	GPS needs clear sky view. Ceramic patch antenna recommended (LCSC C177611).

■ GPS does NOT work indoors without an active antenna. For testing, run the device near a window or outside. Always test outdoors first before assuming the chip is faulty.

8. RTL-SDR v3 — Corrected Architecture

✗ CRITICAL: The original guide described RTL-SDR as integrated on the PCB via ESP32-S3 USB OTG host. This architecture is NOT feasible. The ESP32-S3 does not have enough RAM or CPU to demodulate RTL-SDR IQ streams locally, and ESP32 USB OTG host support for RTL-SDR has no working community implementation. The corrected architecture is below.

Corrected architecture

RTL-SDR v3 is an EXTERNAL USB dongle. It does NOT live on the Flipper Black PCB. It plugs into a laptop or phone via USB. Software (SDR# on Windows, GQRX on Linux/macOS) runs on the laptop and handles all demodulation.

In your Burnout pitch and demo, RTL-SDR is a companion tool — not an onboard feature. Remove RTL-SDR from the integrated BOM. It is a separate \$25–40 dongle the user plugs into their laptop alongside the Flipper Black device.

What this means for your BOM

Remove RTL-SDR from the onboard component list. The rest of the BOM is unaffected. Your total cost drops by \$25–40. If you want SDR capability mentioned in the pitch, phrase it as: 'Compatible with RTL-SDR v3 dongle for full-spectrum scanning via companion laptop software.'

Software	Platform	URL
SDR# (SDRSharp)	Windows	airspy.com/download/
GQRX	Linux / macOS	gqrx.dk/
RTL-SDR Quick Start Guide	All	rtl-sdr.com/rtl-sdr-quick-start-guide/

9. Power Components

BMS — Battery Management System

Part	Function	LCSC #	Notes
TP4056	1A Li-ion charger IC with LED status	C382139	Handles charging from USB-C
DW01A	Li-ion protection: overcharge, overdischarge, short circuit	C351410	Always pair with TP4056
IP5306	Integrated charger + boost + BMS + fuel gauge	C181692	One-chip solution but harder to source as raw chip

✓ TIP: Recommended: use TP4056 for charging + DW01A for protection. Combined cost under \$1. DW01A protects against all three dangerous Li-ion conditions.

LDO — 3.3 V Regulator

Part	Max current	Dropout voltage	LCSC #	Notes
AMS1117-3.3	800 mA	1.1 V at 800 mA	C6186	Most common, SOT-223. High dropout.
AP2112K-3.3	600 mA	0.3 V at 600 mA	C51118	Low dropout, better for battery operation.
TLV75533P	1 A	0.2 V at 1 A	C87990	Best option if total load approaches 1 A — handles RF TX peaks.

Battery selection — 2500–3000 mAh target

Spec	What to look for	Avoid
Capacity	2500–3000 mAh (realistic for this form factor)	Cells claiming >4000 mAh in same physical size — fake
Chemistry	18650 cylindrical or LiPo flat pouch	LiFePO4 (voltage too low for direct use)
Discharge rate	1C minimum (2.5 A for 2500 mAh cell)	Cells rated under 0.5C — will sag under RF TX load
Connector	JST-PH 2.0 or JST-XH 2.54 mm 2-pin	Bare wire terminals without a connector
Supplier	Molicel, Samsung 25R, LG MJ1, Sanyo NCR	Unknown brand 'ultrafire' or similar no-name cells

10. Passives — Resistors, Capacitors, Crystals

Resistors

Value	Use case	Package	LCSC #	Qty
10 kOhm	Pull-ups: CHIP_PU, GPIO0, GPIO46, /WP, /HOLD, SDA, SCL	0402	C98220	15
100 kOhm	Pull-ups: reset buttons, BOOT strap	0402	C25803	5
33 Ohm	SPI series termination — reduces ringing at high speed	0402	C25105	6
1 kOhm	LED current limiting, general purpose	0402	C21190	5
4.7 kOhm	I2C pull-ups for SDA/SCL (PN532 I2C mode)	0402	C25905	4
0 Ohm	PCB jumper / test point bridge	0402	C17168	5

Capacitors

Value	Dielectric	Use case	Package	LCSC #	Qty
100 nF	X7R / X5R	Decoupling: every chip VCC pin	0402	C14663	40
10 pF	C0G / NP0	Crystal load caps (match crystal spec exactly)	0402	C14858	4
1 uF	X5R	Bulk decoupling near chips	0402	C52923	10
10 uF	X5R	Bulk storage near LDO output	0603	C19702	6
4.7 uF	X5R	LDO input/output bypass	0402	C23733	4
100 uF	Electrolytic	Main bulk cap on 3.3 V rail	1206	C45783	2

✗ **CRITICAL:** NEVER use Y5V dielectric capacitors for decoupling. Their capacitance drops by up to 82% at 3.3 V operating voltage — a '100 nF' Y5V cap becomes ~18 nF in circuit. Always use X5R or X7R. Check the dielectric type in the part description before ordering.

Crystal — 40 MHz for ESP32-S3

Spec	Required value	Notes
Frequency	40 MHz exactly	Not 26 MHz, not 32 MHz — must be 40 MHz
Load capacitance	10 pF (most common)	Must match your PCB caps. Use same value.
Package	SMD 3225 (3.2 x 2.5 mm)	Standard for Espressif reference designs
ESR (max)	Under 80 Ohm	High ESR crystals won't start reliably
Frequency stability	+/-20 ppm or better	Affects WiFi channel accuracy
LCSC part	C256661 (Yangxing Tech 40 MHz)	Confirmed compatible with ESP32-S3

11. Where to Buy — Supplier Guide

Supplier	Best for	URL	Notes
LCSC	Raw chips: ESP32-S3, CC1101, SX1276, PN532, passives	lsc.com	Cheapest for raw chips. Works with JLCPCB SMT assembly directly.
JLCPCB	PCB fabrication + SMT assembly	jlcpcb.com	Upload Gerbers + BOM + placement files. Assembles most LCSC parts.
Mouser	Guaranteed genuine parts	mouser.com	More expensive but reliable for critical RF chips.
DigiKey	Wide selection, crystals, passives	digikey.com	Excellent datasheet links from part pages.
AliExpress	RTL-SDR dongle (external), antennas, enclosure	aliexpress.com	Only for non-critical parts. Avoid raw ICs — fakes exist.
Semtech (Mouser)	SX1276 LoRa chip	Search SX1276IMLTRT on mouser.com	Only buy from authorized distributors.
Texas Instruments (DigiKey)	CC1101 sub-GHz chip	Search CC1101RGPR on digikey.com	Genuine TI parts from authorized channels only.

✗ **CRITICAL:** For RF chips (CC1101, SX1276), always buy from authorized distributors (Mouser, DigiKey, LCSC). Counterfeit RF chips are common on AliExpress — they look identical but have completely wrong RF characteristics and will ruin your range and sensitivity.

12. Datasheet Search Strategies

Finding the right datasheet

Best formula: [exact part number] + 'datasheet' in Google. Always prefer the manufacturer's official website over aggregator sites like alldatasheet.com — aggregators often host outdated revisions.

What you need	Search formula	Example
Official datasheet	[PART] site:[manufacturer.com] datasheet	CC1101 site:ti.com datasheet
Application note	[PART] application note site:[manufacturer.com]	CC1101 application note site:ti.com
Errata	[PART] errata site:[manufacturer.com]	ESP32-S3 errata site:espressif.com
LCSC part number	Search exact manufacturer part number on lcsc.com	Search 'W25Q128JVS1Q' not 'W25Q128'
KiCad footprint	Search part number on snapeda.com or ultralibrian.com	Always verify pad dimensions vs datasheet
Reference designs	[CHIP NAME] schematic KiCad on github.com	Flipper Zero hardware is open source — study it

Community help

Forum	URL
ESP32 hardware forum	esp32.com/viewforum.php?f=19
Reddit r/AskElectronics	reddit.com/r/AskElectronics
Reddit r/esp32	reddit.com/r/esp32
EEVblog forum	eevblog.com/forum/
Flipper Zero open source hardware	github.com/flipperdevices/flipperzero-firmware

✓ TIP: When posting for help: always share the exact part number, your schematic snippet, the symptom, and what you have already tried. Vague questions get vague answers.

Master Reference — All Datasheets

Chip	Manufacturer	Primary datasheet URL
ESP32-S3	Espressif	espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf
W25Q128JV	Winbond	winbond.com/resource-files/w25q128jv_dth_rev06_11162016_plus.pdf
CC1101	Texas Instruments	ti.com/lit/ds/symlink/cc1101.pdf
SX1276	Semtech	semtech.com/uploads/documents/DS_SX1276-7-8-9_W_APP_V7.pdf
PN532	NXP Semiconductors	nxp.com/docs/en/user-guide/141520.pdf

Chip	Manufacturer	Primary datasheet URL
NEO-6M	u-blox	content.u-blox.com/sites/default/files/products/documents/NEO-6_HardwareIntegrationManual_(GPS.G6-HW-09007).pdf
DW01A (BMS)	Fortune Semiconductor	datasheet.lcsc.com/szlcsc/Fortune-Semiconductor-DW01A_C351410.pdf
TP4056 (charger)	Various	datasheet.lcsc.com/szlcsc/UMW-TP4056_C382139.pdf
AP2112K-3.3 (LDO)	Diodes Inc.	diodes.com/assets/Datasheets/AP2112.pdf
40 MHz crystal	Yangxing Tech	datasheet.lcsc.com/lcsc/2208061800_Yangxing-Tech-X322540MSB4SI_C256661.pdf